

Practice Quiz 10C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Images and inverse images; the geometric sum formula.

1. (a) Let $A = \{1, 2, 3, 4, 5\}$, $B = \{p, q, s\}$, and define

$$f(1) = p, \quad f(2) = q, \quad f(3) = p, \quad f(4) = s, \quad f(5) = q.$$

For $X = \{2, 4\}$, compute $f(X)$ and $f^{-1}(f(X))$. Are the latter and X equal?

Solution $f(X) = \{q, s\}$. The elements mapped into $\{q, s\}$ are 2, 4, 5, so

$$f^{-1}(f(X)) = \{2, 4, 5\} \neq \{2, 4\} = X.$$

The extra element 5 appears because $f(5) = f(2) = q$. ■

- (b) For real $r \neq 1$, state and prove by induction a closed formula for

$$1 + r + r^2 + \cdots + r^{n-1} \quad (n \geq 1).$$

Solution The formula is

$$\sum_{i=0}^{n-1} r^i = \frac{r^n - 1}{r - 1}.$$

For $n = 1$, both sides equal 1. Suppose the formula holds for an integer $n \geq 1$. Then

$$\begin{aligned} \sum_{i=0}^n r^i &= \frac{r^n - 1}{r - 1} + r^n \\ &= \frac{r^n - 1 + r^n(r - 1)}{r - 1} = \frac{r^{n+1} - 1}{r - 1}. \end{aligned}$$

This is the formula for $n + 1$, so induction proves the result. ■